THESIS

Propose. Prove. Execute.

An options agent that must show its work.

Thesis is an autonomous options agent that cannot risk a dollar until it writes why the trade exists, when it is wrong, and the maximum amount it can lose. Grok proposes a falsifiable idea; deterministic code decides whether it is tradable; Alpaca paper infrastructure proves what occurred.

AI LOGIC

Creative thesis. Bounded authority.

Each cycle reads live Alpaca paper-account state, positions, fills, and market data. Grok-4.6 returns structured JSON: underlying, direction, regime, setup, invalidation, horizon, expected move, IV note, and conviction. It never chooses contracts, sizes risk, or submits orders.

At qualifying conviction, code constructs a call or put debit vertical: 14–45 DTE near 25 DTE, approximately at-the-money long leg, defined-width short leg, and 1:1 day-limit MLEG. Maximum loss is known before submission: net debit × 100 × quantity.

RISK GATES

Language cannot authorize.

Hard controls require a paper endpoint; active, unblocked account; options level 3; valid CLI clock agreeing with the SDK; allowlisted underlying; conviction ≥0.35; valid quotes and debit; 14–45 DTE; market open; no more than three open theses; ≤2% equity risk per thesis; ≤6% aggregate risk; and explicit execution enablement.

Any failure records a no-trade or blocked decision. Abstention is evidence, not an error.

ALPACA INFRASTRUCTURE

Tool evidence, not a screenshot.

Alpaca Trading and Market Data APIs supply account state, contracts, quotes, bars, orders, fills, positions, and portfolio history. Every cycle runs `alpaca account get` and `alpaca clock`. Eligible orders submit only through `alpaca api POST /v2/orders`.

Invalid CLI JSON, account restrictions, clock disagreement, timeout, rejection, or a response without an order ID stops the cycle. There is no SDK order fallback after an ambiguous result.

EVIDENCE & PERFORMANCE

The read-only FastAPI console reconciles an append-only SQLite decision ledger with Alpaca orders, fills, positions, and portfolio history. Judges can inspect the thesis, gates, sanitized CLI/API trace, MLEG intent, broker state, realized and unrealized P&L, reconciliation delta, and equity curve. Values come directly from the submitted paper account; none are hardcoded.

WHY THESIS

Evidence over assertion. The model is creative where language is useful and powerless where deterministic controls are safer. Every trade—and every refusal to trade—is explainable from one screen.